HyperFitPro Studio - User Guide

First public release help manual

1. Purpose

HyperFitPro Studio is an engineering workbench for hyperelastic model calibration, optimization, reporting, plugin diagnostics and FEA export.

2. Guided workflow

1) Working directory, 2) model selection, 3) data import, 4) optimization, 5) report/export, 6) plugin analysis, 7) .hyp2fit project save.

3. Worked examples

Example 1: Neo-Hookean quick calibration. Example 2: Mooney-Rivlin multi-mode fit. Example 3: Yeoh force-displacement workflow. Example 4: Ogden global optimization. Example 5: Gent / Van der Waals plus FEA export verification.

4. Plugin Workbench

Fifty embedded tools are grouped by category. They can be run on the active workspace or imported projects. Output files are listed in Plugin Results.

5. Project Library

Saved .hyp2fit projects can be imported, reloaded and used as plugin targets. This supports project-level post-processing independent of the active run.

6. Languages

The header language selector switches the main GUI among Turkish, English, German, French and Spanish.

7. Verification

Use Diagnostics Center before committing to GitHub. It runs core, model library, data processing, optimization, FEA export and plugin checks.

Example data files

Example	Model	Data type
1	Neo-Hookean	uniaxial + biaxial
2	Mooney-Rivlin	uniaxial + biaxial + planar
3	Yeoh	force-displacement
4	Ogden	high-strain uniaxial
5	Gent / Van der Waals	finite extensibility + volumetric